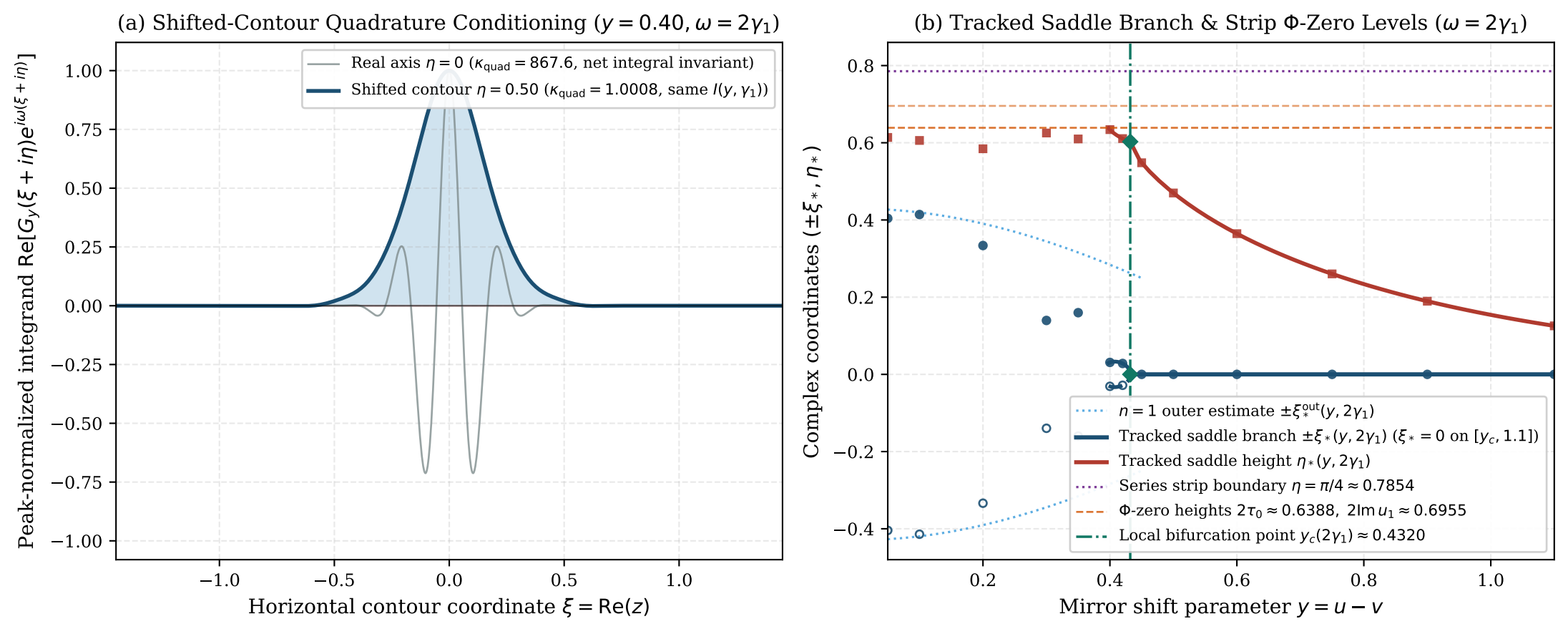

(a) Shifted-Contour Quadrature Conditioning ($y = 0.40, \omega = 2\gamma_1$)
Real axis $\eta = 0$ ($\kappa_{\text{quad}} = 867.6$, net integral invariant)
Shifted contour $\eta = 0.50$ ($\kappa_{\text{quad}} = 1.0008$, same $I(y, \gamma_1)$)
Peak-normalized integrand $\mathrm{Re}[G_y(\xi + i\eta)e^{i\omega(\xi+i\eta)}]$
Horizontal contour coordinate $\xi = \mathrm{Re}(z)$
(b) Tracked Saddle Branch & Strip Φ-Zero Levels ($\omega = 2\gamma_1$)
Complex coordinates $(\pm\xi_*, \eta_*)$
Mirror shift parameter $y = u - v$
$n = 1$ outer estimate $\pm\xi_*^{\text{out}}(y, 2\gamma_1)$
Tracked saddle branch $\pm\xi_*(y, 2\gamma_1)$ ($\xi_* = 0$ on $[y_c, 1.1]$)
Tracked saddle height $\eta_*(y, 2\gamma_1)$
Series strip boundary $\eta = \pi/4 \approx 0.7854$
Φ-zero heights $2\tau_0 \approx 0.6388$, $2\mathrm{Im}\, u_1 \approx 0.6955$
Local bifurcation point $y_c(2\gamma_1) \approx 0.4320$